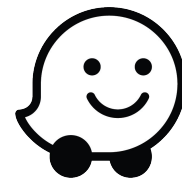


# Waze Churn Project | Machine Learning Model



## Executive Summary

### ISSUE / PROBLEM

At the final stage of the project we're required to develop a machine learning classification model that have reliable performance using data that is not equally balanced and have much higher percentage of retained users.

### RESPONSE

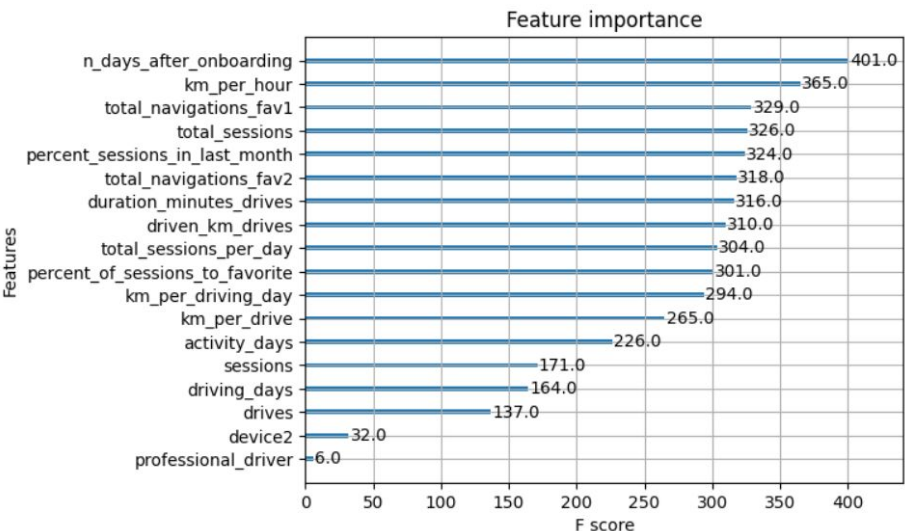
The data were splitted into 60% training, 20% validation and 20% testing data where each set stratify the distribution of the original data classes. Random forest and xgboost models were developed in tuning parameters technique to yield the model with higher recall score (i.e. lower false negative errors) considering the goal of detecting user churn tendency.

### IMPACT

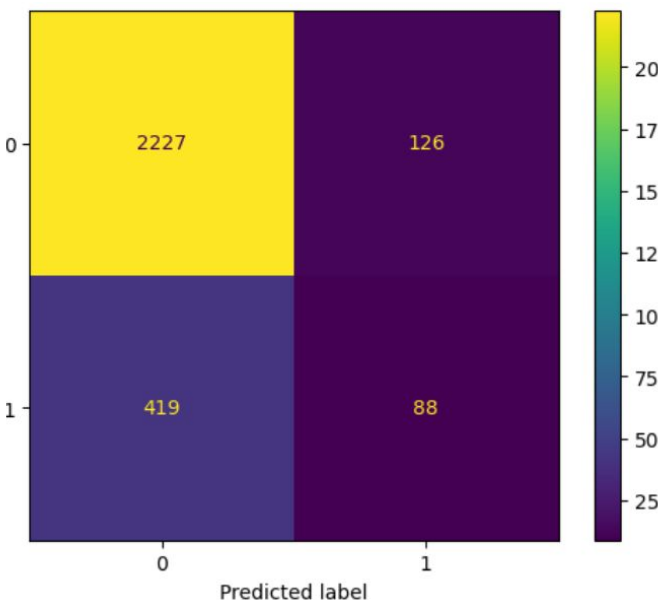
Xgboost model was selected as the champion where it showed relatively good scores on the test data compared to logistic regression and random forest.

| precision | recall  | F1       | accuracy |
|-----------|---------|----------|----------|
| 0.411215  | 0.17357 | 0.244105 | 0.809441 |

Figure shows the importance of different features on model's prediction



Confusion matrix shows the model high tendency of making false negative errors (where churned = 1, retained = 0)



### KEY INSIGHTS

- We don't recommend using the current model for churn prediction, because it yielded many false negative errors (where it classify users who will churn as retained). However the model will be of huge benefit for guiding future exploratory analysis efforts.
- In order to develop the model performance, more data could be collected that need to balance the classes in the target variable to reveal more about churn customer behavior. Also more future engineering could be done to improve the model since it is one of the easiest and fastest ways to improve model performance, where we can see most variables that are considered important for prediction were gained by feature engineering.